COMP8851 - MLP Baseline Report

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1. Model Summary

Parameter	Value
Model	Multi-Layer Perceptron (MLP)
Input	Flattened 200x9 window (1800 features)
Output	sin(heading) and cos(heading)
Architecture	Linear(1800,512) > ReLU > Dropout(0.3) > Linear(512,256) > ReLU > Dropout(0.3) > Linear(256,128) > ReLU > Linear(128,2)
Total Parameters	1,086,594
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Early Stopping	Patience 5, triggered at epoch 24
Best Val Loss	0.369177
Training Time	167.33 seconds (2.79 minutes)
Device	NVIDIA GeForce RTX 4070 Laptop GPU

2. Quantitative Results

Metric	Value
MAE on test set (degrees)	68.7472°
RMSE on test set (degrees)	85.3652°
Training time	167.33 seconds (2.79 minutes)

3. Scenario Breakdown

Scenario	MAE (degrees)
Straight walking	62.6268°
Sharp turns	88.7315°
Short trajectories	65.6208°
Long trajectories	70.8040°

4. Full Model Comparison

Model	MAE (°)	RMSE (°)
EKF Baseline	86.79	102.13

MLP (this model)	68.75	85.37
TCN	69.27	85.99
LSTM Original	67.05	83.61
LSTM Improved	69.22	86.08
Transformer Original	66.99	83.30
Transformer Improved	66.65	82.44
Hybrid Original	66.31	82.99
Hybrid Improved	66.77	83.53

5. Reflection Part 1 - Individual

Q1. Does MLP outperform EKF and why?

Yes. MLP achieved MAE 68.75° compared to EKF 86.79°, an improvement of 18.04°. The EKF is a mathematical filter that relies on magnetometer readings which are systematically distorted in indoor environments, and its linearisation of nonlinear heading dynamics via first-order Taylor expansion fails during rapid directional changes. The MLP on the other hand learns directly from raw sensor patterns in the training data without any handcrafted assumptions. Even without temporal structure, the MLP can learn statistical associations between sensor channel combinations and heading direction that the EKF cannot capture through its fixed mathematical model. This confirms that even the simplest learned approach significantly outperforms a traditional filter-based method on this task.

Q2. How close is MLP to LSTM and Transformer and what does this tell us?

MLP at 68.75° MAE sits approximately 1.7° above the original LSTM (67.05°) and 1.76° above the original Transformer (66.99°). This gap is relatively small - under 2 degrees - which is a meaningful finding. It suggests that while temporal modelling does provide a measurable benefit, the improvement is modest rather than dramatic. The bulk of the performance gain over EKF comes simply from learning sensor-to-heading associations from data, not specifically from modelling temporal dependencies. This implies that the generalisation bottleneck for this task is primarily driven by subject variability across the train and test splits rather than architectural sophistication.

Q3. What does the loss curve tell us about training behaviour?

The training loss decreased consistently across 24 epochs from 0.455 to approximately 0.343 before early stopping. The validation loss also improved overall but showed more fluctuation, plateauing around epoch 19 before early stopping triggered at epoch 24. The gap between training and validation loss widened gradually over training, indicating mild overfitting. The model learned the training subject patterns reasonably well but struggled to generalise fully to unseen subjects. This is consistent with what was observed across all other models in this project - subject generalisation appears to be the dominant challenge regardless of architecture.

Q4. Where does MLP struggle most and why?

MLP struggles most on sharp turns with MAE 88.73° and long trajectories with MAE 70.80°. The sharp turn weakness is directly attributable to the lack of temporal structure. When a pedestrian makes a sharp turn, the heading changes rapidly across consecutive timesteps. An MLP flattens the entire 200-timestep window into a single feature vector and loses all information about the ordering and progression of sensor readings over time. It cannot track the temporal dynamics of a directional transition the way LSTM or Transformer can. Long trajectory weakness likely reflects accumulated subject-specific drift that the model cannot compensate for without temporal context. Straight walking at 62.63° is the easiest scenario because heading is stable and the static sensor patterns are more consistent and learnable without temporal awareness.

6. Reflection Part 2 - Cross-Model

Q1. How does MLP rank against all models including improved versions?

MLP at 68.75° MAE ranks fourth overall out of nine model configurations. It outperforms EKF (86.79°), TCN (69.27°), and notably the improved LSTM (69.22°). It sits below the original LSTM (67.05°), original Transformer (66.99°), improved Transformer (66.65°), original Hybrid (66.31°), and improved Hybrid (66.77°). The fact that MLP outperforms the improved LSTM is a notable finding - it suggests that the LSTM improvement experiments, which increased dropout and window size, actually hurt overall generalisation performance. This tells us that temporal modelling capacity alone does not guarantee better performance and that training configuration choices matter as much as architectural complexity.

Q2. Does adding MLP strengthen or weaken the argument for deep learning?

Adding MLP significantly strengthens the argument for deep learning over filter-based methods. The EKF to MLP jump of 18.04° MAE demonstrates that learned approaches are fundamentally superior to handcrafted filter-based methods for indoor heading estimation. However MLP also reveals that the performance gains from adding temporal structure - LSTM and Transformer - are relatively modest at around 1-2 degrees MAE. This nuances the narrative: deep learning decisively beats traditional methods, but the specific type of temporal architecture matters less than expected, with subject generalisation being the primary bottleneck across all learned models. The MLP baseline therefore adds important context to the final comparison by isolating the contribution of temporal modelling from the contribution of learning itself.